VINEETH KALLURU

Applied AI Engineer | Production LLM Systems, Agents and Post-Training | IEEE Senior Member

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SUMMARY

Applied AI engineer who turns open models and agents into production systems for enterprises. Three years at NVIDIA as the technical lead for strategic software partners, spanning solution architecture, post-training, evaluation, inference and executive advisory. Earlier built ML hardware and compilers at Google Cloud TPU and SambaNova. Publishes on agentic AI reliability and serves on conference program committees.

EXPERIENCE

Senior Solutions Architect, Generative AI, NVIDIA **Nov 2023 - 2026**

- **Production AI delivery:** Technical lead for strategic enterprise software partners, taking open models and agentic systems from proof of concept to production and general availability.
- **Solution architecture:** Designed RAG, agentic and guardrailed AI systems and the GPU infrastructure behind them across cloud, on-prem and air-gapped environments.
- **Post-training and evaluation:** Shaped post-training data strategy and evaluation methodology with partner ML teams, and turned field failure analysis into improvements in NVIDIA's own models.
- **Inference performance:** Architected large-scale LLM serving and led NVIDIA's technical support for a hyperscale cloud partner's MLPerf Inference submissions.
- **Strategic advisory:** Trusted technical advisor to partner product and engineering leadership; led the technical track of executive briefings and co-owned joint roadmaps and launch announcements.
- **Product influence:** Converted field issues into product fixes, roadmap items and process changes across NVIDIA's model, retrieval, guardrails and inference products.
- **Reusable assets:** Authored the Nemotron-CC data-curation pipeline in NeMo Curator, adopted by external foundation-model teams, and built a multi-agent reference workflow taught as a GTC workshop.
- **Team multiplier:** Mentored solutions architects and interns, led account transitions, built cloud GTM enablement content, and interviewed for SA hiring.

ML Accelerator Architect, Cloud TPU, Google **Jul 2022 - Nov 2023**

- Performance analysis of Gemini workloads on TPU v6 and v7 compute and memory subsystems; hardware-software co-design under Norm Jouppi and David Patterson.

Compiler / ML Systems Engineer (Team Lead), SambaNova Systems **Feb 2020 - Jul 2022**

- Led a 7-engineer team that shipped the first known production for-loop and RNN support in a dataflow compiler: about 4x faster online inference and 5x faster LSTM training vs. NVIDIA A100 (SambaNova public blog, Aug 2022); about 3x chip-utilization gain from BERT tiling and compiler work.

SELECTED PROJECTS

- **rl-for-ai-research:** GRPO reinforcement-learning training on FinanceBench; open source.
- **WG-Plan:** Workflow-grounded small-model planners for domain-specific tool calling, using workflow graphs to constrain valid actions (research in progress).
- **jarvis-desk-bot:** Hermes agent voice front end on an ESP32 desk robot (local speech, spoken approvals); MIT licensed.

SELECTED PUBLICATIONS

- **Between Weights and Workflows: Executable Graph Contracts for Bounded Enterprise Agents.** IEEE ICA 2026. Accepted (oral); first author.
- **The Coherence Trap: Layer-Decomposed Failure-Mode Discovery in a Deployed Agentic LLM System.** IEEE ISSRE 2026 Industry Track (CORE A). Presented; first author.
- **Beyond the Blue Link** (generative engine optimization in stochastic retrieval) and **Edit-Distance Links** (conditional-tokenization world models). ACM Hypertext 2026; published.
- **NVIDIA Nemotron technical reports:** Nemotron Nano 2 (arXiv:2508.14444, co-author); Llama-Embed-Nemotron-8B (arXiv:2511.07025, contributor).
- **FinanceHarness: Engineering a Finance Retrieval Harness for Search Agents Powered by Sub-10B Models.** RL4LLM-Agents workshop, ACM ICAIF 2026. Accepted; sole author.
- **SkillOpt: Trajectory-Derived, Verifier-Grounded Compilation of LLM-Agent Skills.** ACM CAIS 2026 Agent Skills Workshop; 63% fewer tokens and 62% lower latency at non-regressing reward.
- **HANNA: Hardware Aware Neural Network Architectures using FbNet.** arXiv:1906.07214; first author.

SPEAKING

- NVIDIA GTC 2026 full-day DLI workshop (co-instructor) and poster; GTC DC 2025; Microsoft Build 2024 and Ignite 2024; SC25 demo; SEMI Agentic AI for Semiconductor Manufacturing 2025.
- O'Reilly AI Superstream 2026 ("Building Agent Harnesses to Match Frontier-Model Performance in RAG"); AIKP 2026 invited workshop; TechEquity AI Summit 2026 (Nov 5); CMU guest lecturer (18-740).

RECOGNITION

- **IEEE Senior Member**; IEEE-HKN; NVIDIA NIM Coin nomination and GTC 2026 Spot Award; two granted Indian patents in applied AI/ML (IN 431037, IN 451829); winner, Apple-sponsored Memory Fairness Protocol competition (CMU).

PEER REVIEW (CONFERENCES AND WORKSHOPS)

- **Program committee:** IEEE ICA 2026; KDD 2026 AgenticSE and Agent4IR workshops.
- **Reviewer:** NeurIPS 2026 (ethics review; UserSim and Continual World Models workshops); ACM CAIS 2026 Agent Skills workshop; KDD 2026 workshops (Enterprise AI Agents, AI Data Scientist, SciSoc Agents and LLMs, RelKD); IEEE DSAA 2026; AIKP 2026; IEC2T 2026.

SKILLS

Inference: TensorRT-LLM, NVIDIA Dynamo, NIM, vLLM, Run:ai, disaggregated and multi-node serving, benchmarking (MLPerf)
Models and agents: Nemotron and open LLMs, post-training data design, GRPO, RAG, agent harnesses, NeMo Curator, NeMo Guardrails, NeMo Agent Toolkit, LLM evaluation
Systems and cloud: Python, C/C++, CUDA, ML compilers, Kubernetes; Azure (AzureML, AI Foundry, AKS), AWS, GCP; on-prem and air-gapped

EDUCATION

M.S., Electrical and Computer Engineering, Carnegie Mellon University — **Aug 2018 - Dec 2019**

Professional Program in Artificial Intelligence, Stanford University — **Sep 2022 - Dec 2022**

B.Tech., Electronics and Communication Engineering, SRM Institute of Science and Technology — **Jul 2014 - Apr 2018**